

IT Workshop-I (CS 202)

Assignment 1

Total Marks =20

Q1. A teacher takes attendance during two different lab sessions. The roll numbers of students present in each session are stored in sorted arrays. The teacher wants to find the students who attended **both sessions**. Write a method that returns an array containing the common roll numbers. **[Marks =10]**

Test Case:

Morning Session: {2, 5, 8, 12, 15, 20}

Afternoon Session: {3, 5, 8, 10, 15, 18}

Output:

Students Present in Both: {5, 8, 15}

Q2. An electricity board stores the daily electricity consumption of a household in an array. Write a method to find the length of the longest continuous period during which the electricity consumption keeps increasing every day. Solution should be in $O(n)$ time complexity and $O(1)$ space complexity **[Marks =7]**

Input: {120, 135, 150, 142, 160, 175, 190, 185, 200}

Output: Longest Increasing Streak: 4

Here, 142, 160, 175, 190 forms the longest increasing streak.

Note: Marks =3 for good programming practice